

FAISAL AHMED YOUSIF AL-DROU

Junior Machine Learning Engineer

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PROFESSIONAL SUMMARY

Artificial Intelligence and Robotics Engineer with practical experience in developing machine learning models, data processing pipelines, and deep learning solutions using Python and modern AI frameworks. Experienced in applied AI projects including forecasting, data analysis, and image classification, with additional experience in computer science and robotics training. Ranked 2nd academically, with the ability to deliver data-driven AI solutions that enhance operational efficiency and support smarter decision-making for the organization.

TECHNICAL SKILLS

Artificial Intelligence & Generative AI:	Machine Learning, Deep Learning, Neural Networks, LangChain Expression Language (LCEL), Multi-Provider Model Routing, Prompt Engineering, Predictive Modeling, RAG & Agentic AI (In Progress)
AI Frameworks & Libraries:	LangChain, Pydantic v2, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna
Data Science & Output Parsing:	Pandas, NumPy, JSON Output Parsing, Feature Engineering, Data Preprocessing, Data Transformation, Data Analysis, Apache Parquet
Web & Middleware Architecture:	Python (Flask), RESTful API Design, Middleware Architecture, Error Handling & Fallback Pipelines, Latency Tracking & Benchmarking
Programming Languages:	Python, SQL, C++, JavaScript (Fetch API / Async), Kotlin, Arduino C
Development Tools & Security:	Git, Linux, Bash, Ubuntu Command Line, Environment Variable Management (python-dotenv), Secret Isolation

PROFESSIONAL EXPERIENCE

Computer Science & Robotics Instructor	Amman, Jordan
Al-Namothaj School for Special Education	
Delivered practical training in computer science and robotics, enabling students to understand programming concepts through hands-on activities.	
- Developed simplified learning materials for complex technical concepts, improving students' understanding and application of programming skills. - Designed interactive technical activities that enhanced students' problem-solving abilities and computational thinking skills.	

TECHNICAL PROJECTS

Multi-Provider GenAI Support Application
Technologies: Python (Flask), LangChain (LCEL), Pydantic v2, REST API, Google Gemini, Meta Llama, JavaScript
Engineered a multi-provider AI support application featuring a unified middleware interface that abstracts model execution across Google Gemini and Meta Llama.
- Implemented deterministic JSON parsing using Pydantic schemas and LangChain's JsonOutputParser to enforce structured output data for backend consumption. - Built defensive error-handling logic and fallback pipelines to manage API rate limits and prevent server crashes during service degradation. - Integrated asynchronous JavaScript Fetch API for non-blocking UI operations and implemented real-time latency tracking across model providers.
Seasonal Temperature Forecasting Pipeline
Technologies: Python, LightGBM, CatBoost, XGBoost, Optuna, Scikit-learn
Developed a machine learning model for seasonal temperature forecasting, enabling analysis of climate patterns and generating data-driven future predictions.

• **New York City Collision Data ETL Pipeline**

Technologies: Python, Pandas, NumPy, JSON, Apache Parquet

Built a data processing pipeline to clean, transform, and organize large-scale collision datasets, improving data quality and preparing it for analysis and insight extraction.

• **Binary Image Classification Using Convolutional Neural Networks (CNN)**

Technologies: Python, TensorFlow, Keras, NumPy, Matplotlib

Developed a deep learning model for automated image classification, enabling pattern recognition through convolutional neural network techniques.

EDUCATION

Artificial Intelligence and Robotics Engineering

Graduation Date: 2024

GPA: 83.3% | **Academic Rank:** Ranked 2nd in class

TECHNICAL TRAINING & CERTIFICATIONS

- **IBM RAG and Agentic AI Professional Certificate (In Progress)** – Coursera / IBM
 - Completed: "Develop Generative AI Applications: Get Started" (LangChain, LCEL, Pydantic, Flask)
- **Data Science and Artificial Intelligence Training** – SHAI for AI
- **Practical Training Program** – Infotel
- **Artificial Intelligence and Machine Learning in Medical Applications** – Course The Hope International
- **Python for Machine Learning Certificate** – Pro Vision

LANGUAGES

Arabic: Native

English: Advanced

Japanese: Intermediate